The Journal of Open Source Software

OPTIMUS: a multidimensional global optimization package

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DOI: 10.xxxxxx/draft

Software

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Submitted: 14 October 2024
Published: unpublished

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Summary

The location of global minima is applied in a variety of problems derived from physics (Duan et al., 1992), chemistry (Liwo et al., 1999), medicine (Lee, 2007) etc. Also, global optimization can be used in complex problems, such as the training of neural networks (Chiroma et al., 2017). Methods used in the global optimization process can be divided into stochastic and deterministic and a comparison between methods of these categories can be found in the work of Sergeyev et al.(Sergeyev et al., 2018). This article proposes a software that incorporates a variety of stochastic methods to handle optimization methods of multidimensional functions.

Statement of need

This paper presents the `GlobalOptimus` software, written in ANSI C++, which allows researchers wanting to solve optimization problems in their domain to formulate their optimization problems and apply different global optimization methods to identify the global minimum of the problem. The main features of the proposed software are: a) Coding of the objective problem in a high level language such as ANSI C++ b) Incorporation of many global optimization techniques to solve the objective problem c) Parameterization of global optimization methods using user-defined parameters d) Usage of a GUI application to control the optimization strategy. Some of the frequently used optimization tools are MATLAB Optimization Toolbox, SciPy optimization module and Gurobi. MATLAB and SciPy are usually based on high-level languages, which can limit performance when computational problems are required. GlobalOptimus is an algorithmic optimization tool written in C++ that combines high performance and ease of parameterization for solving optimization problems in many domains. While other optimization platforms exist, GlobalOptimus stands out for its openness and customizability.GlobalOptimus differs from other optimization tools in that it offers full support for using various optimization algorithms, such as genetic algorithms, differential evolution, and others. Unlike tools like MATLAB or SciPy, GlobalOptimus allows users to work with an environment that is fully customizable and integrate software from other platforms without restrictions. The fact that it is open source enhances its modifiability. Closing Gurobi which focuses on mathematical optimization requires a commercial license, while GlobalOptimus is open source.

Software

The software can be compiled using the QT library, available from https://qt.io. The researcher should code their objective function and a number of other mandatory functions in the C++ programming language, specifying the dimension of the objective function and the bound of

the function domain. Subsequently, the user can select a global optimization method to apply to the problem from a wide range of available methods.

Implemented global optimization methods

In the proposed software, each implemented global optimization method has a set of parameters that can determine the overall optimization path and the effectiveness of the method. Some of the global optimization methods are as follows:

METHOD	DESCRIPTION
Differential Evolution	An evolutionary algorithm used in many areas (Pant et al., 2020)
Parallel Differential Evolution	A parallel version of the Differential Evolution method (Charilogis & Tsoulos, 2023).
Double precision genetic algorithm	An Improved version of the Genetic Algorithm (Tsoulos, 2008).
Improved Particle Swarm Optimization	Suggested as an improvement of the PSO method (Charilogis & Tsoulos, 2022).
Multistart	Initiates local searches from different start points (Martí et al., 2010).
NeuralMinimizer	Utilizes Radial Basis Functions (RBF) networks to increase the speed of the Multistart method (Tsoulos et al., 2023).
Parallel Particle Swarm optimizer	A novell parallel PSO variant, published recently (Charilogis et al., 2023).
Simulated annealing optimizer	included in the software under the name Simman (Kirkpatrick et al., 1983).
The optimal foraging algorithm (OFA)	motivated by animal behavioral ecology included the software named Ofa (Kyrou et al., 2024a).
Bio-inspired metaheuristic algorithm Giant Armadillo Optimization (GAO)	This process mimics the natural behavior of the giant armadillo (Kyrou et al., 2024b).
The Gray Wolf Optimizer (GWO)	This process mimics hunting mechanism of gray wolves in nature (Li et al., 2021).

Implemented local optimization methods

The parameter `--opt_localsearch` is used to select the used local optimization procedure. The implemented local optimization methods are the following:

METHOD	DESCRIPTION
The bfgs method	The Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm was implemented (Powell, 1989).
The lbfgs method	The limited memory BFGS method is implemented as an approximation of the BFGS method using a limited amount of computer memory (Liu & Nocedal, 1989).
The Gradient descent method	This method is denoted as gradient in the software and implements the Gradient Descent local optimization procedure (Amari, 1993).

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METHOD	DESCRIPTION
The Nelder Mead method	The Nelder Mead simplex procedure for local optimization is also included in the software (Olsson & Nelson, 1975).
The adam method	The adam optimizer (Kingma, 2014).

Implementing a user - defined optimization method

The software can be extended by implementing custom optimization techniques by extending the optimization method class `UserMethod` and implementing the provided functions according to the requirements of the method.

Method	DESCRIPTION
init()	This function is called every time the optimization method starts
step()	This function implements the actual step of the optimization method.
terminated()	This function is used as the termination step of the optimization method.
done()	This function will be called when the optimization method terminates.
~UserMethod()	This is the destructor of the optimization method.

A flowchart of any used optimization process is outlined in Figure Figure 1.

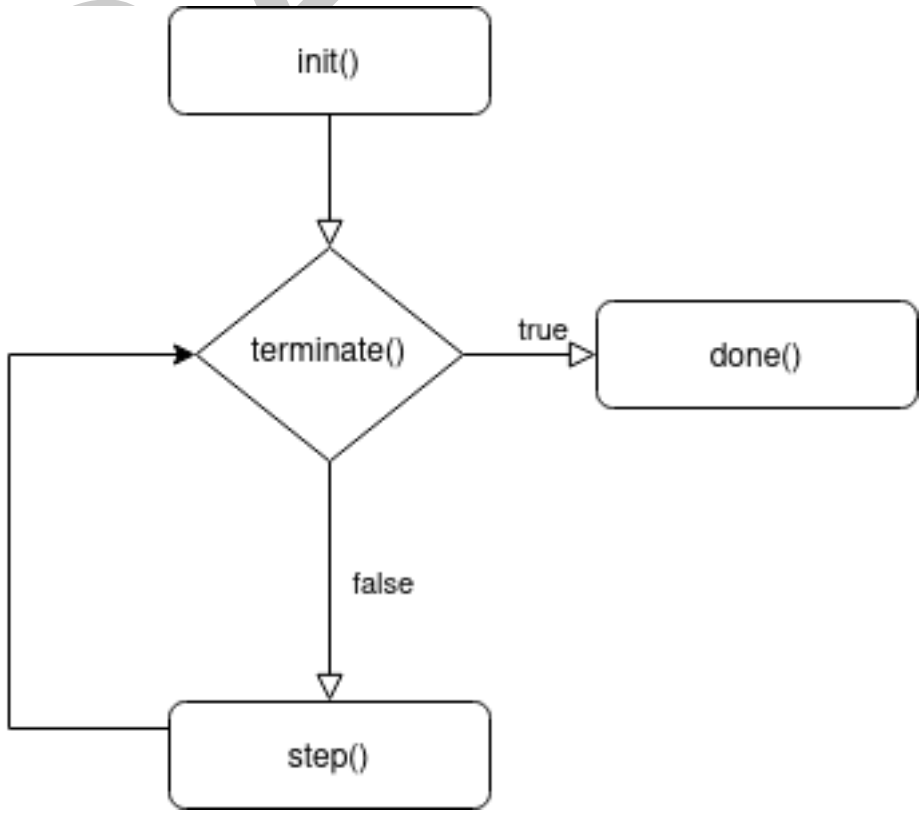

Figure 1: The diagram of the optimization process

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Also, an example run for the function rastrigin and the Genetic algorithm is outlined in Figure Figure 2.

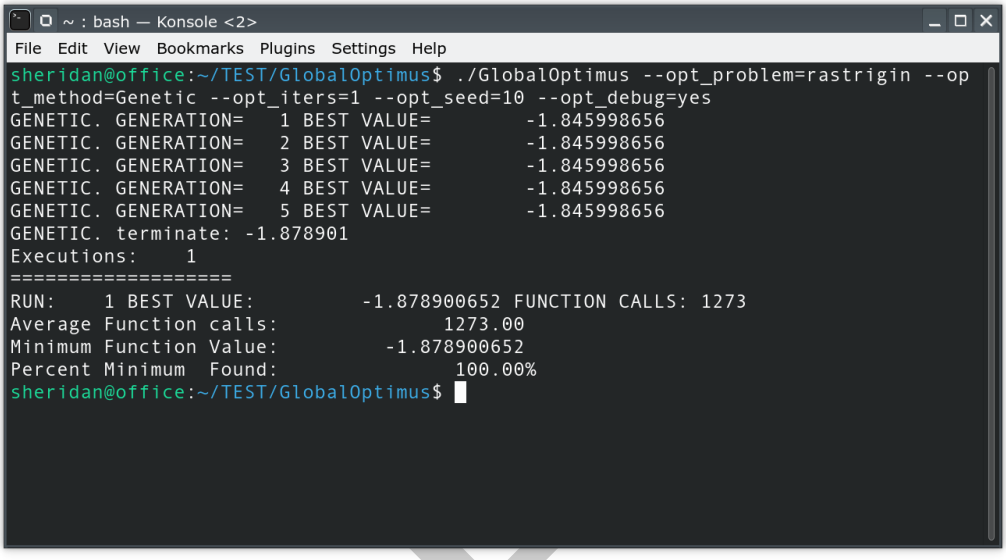

Figure 2: An example run for the function rastrigin

Conclusions

In this work, an environment for executing global optimization problems was presented, where the user can implement their optimization problem using a set of predefined functions and then has the possibility to choose among several global optimization methods to solve the defined problem. In addition, it is given the possibility to choose to use some local optimization method to enhance the reliability of the produced results. This programming environment is freely available and easy to extend to accommodate more global optimization techniques.

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